

Jared Fernandez

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EDUCATION

Carnegie Mellon University: School of Computer Science 2020 – 2026

Ph.D. Candidate in Computer Science, Language and Information Technology

Dissertation: Towards Efficient and Sustainable Machine Learning

Advisors: Emma Strubell & Yonatan Bisk

Northwestern University, *Magna cum Laude* 2015 – 2019

B.S. in Computer Science with Honors & B.S. in Electrical Engineering

Advisors: Doug Downey & Thrason Pappas

RESEARCH EXPERIENCE

Postdoctoral Research Fellow 2026 - 2028

Cornell Tech

New York, NY

Visiting Researcher 2024

Meta, Fundamental AI Research (FAIR)

New York, NY

- Investigated the performance and efficiency scaling properties of large-scale distributed training of neural networks in the multi-thousand GPU regime.
- Identified parallelism techniques for alleviating communication bottlenecks in at-scale training which increased throughput by over 50% (*TMLR 2025*)

Visiting Researcher 2023

Allen Institute for Artificial Intelligence

Remote

- Assisted in creation and implementation of efficiency benchmark for large language models across NLP tasks.

Graduate Student Researcher 2020 – Present

Carnegie Mellon University, Language Technologies Institute

Pittsburgh, PA

- Identified efficiency and latency bottlenecks in software frameworks designed for ML inference (*EMNLP '23*).
- Analyzed the impact of algorithmic & system design decisions on energy use during LLM inference (*ACL '25*)
- Proposed life cycle assessment framework for ML models to evaluate environmental impact of models across development and deployment (*CCAI '25, ICML '26*)
- Created continual training techniques for updating LLMs on new knowledge to reduce need for retraining (*EMNLP Findings '24*)

Visiting Researcher 2022

Google Research, Perception

Mountain View, CA

- Investigated methods for cotraining Vision Transformer models to use fewer tokens for inference time speedups

Undergraduate Researcher 2018 – 2019

Northwestern University, Pappas Group

Evanston, IL

- Designed efficient metrics for measuring perceptual texture similarity for image retrieval and segmentation

Undergraduate Researcher 2017 – 2019

Northwestern University, WebSAIL Group

Evanston, IL

- Designed efficient metrics for perceptual similarity of image texture for image retrieval and segmentation
- Designed methods for building commonsense datasets through adversarial authoring & data augmentation
- Proposed weighted importance sampling techniques to select training data for RNN language models

Research Intern Summer 2013, 2014

Stanford University: School of Earth Sciences, Cosmogenic Radiation Laboratory

Stanford, CA

- Determined slip rates on the San Andreas Fault using mass spectrometry dating of Beryllium isotopes

INDUSTRY EXPERIENCE

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| Software Engineer | 2019 – 2020 |
| <i>Google, Search Notifications</i> | <i>Mountain View, CA</i> |
| <ul style="list-style-type: none">• Pipelined and evaluated the effectiveness of various features for predicting user engagement with targeted Search notifications through experiments on live traffic• Designed infrastructure to predict user engagement with event-based notification subscriptions | |
| Software Engineering Intern | Summer 2019 |
| <i>Google, Dialogflow NLU</i> | <i>Mountain View, CA</i> |
| <ul style="list-style-type: none">• Improved intent recognition & out-of-domain detection models for user conversations with enterprise chat bots• Designed pipeline for domain adaptation and finetuning of sentence embedding models. Increased downstream intent recognition by +4.8% acc. | |
| Software Engineering Intern | Summer 2018 |
| <i>Google, Assistant on Android Auto</i> | <i>Mountain View, CA</i> |
| <ul style="list-style-type: none">• Improved query processing in Google Assistant and query handling by Search.• Built new interfaces for submitting offline queries to improve discoverability for Assistant on Auto. | |

FELLOWSHIPS & AWARDS

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| Empire AI Postdoctoral Fellowship | 2026 – 2028 |
| Christine Mirzayan Science & Technology Policy Fellowship, <i>Finalist</i> | 2026 |
| NSF Graduate Research Fellowship | 2021 – 2024 |
| Carnegie Mellon University Research Fellowship | 2020 – 2021 |
| Tau Beta Pi, <i>Northwestern University</i> | 2018 |
| Eta Kappa Nu, <i>Northwestern University</i> | 2018 |

PEER-REVIEWED PUBLICATIONS

- Evaluation of ML Resource Utilization Requires Model Life Cycle Assessment**
Jared Fernandez, Clara Na, Yonatan Bisk, Constantine Samaras, Emma Strubell
International Conference on Machine Learning: Position Paper Track (ICML), 2026.
- Empirically-Calibrated H100 Node Power Models for Accurate AI Training Energy Estimation**
Alex C. Newkirk, **Jared Fernandez**, Jonathan Koomey, Imran Latif, Emma Strubell, Arman Shehabi, Constantine Samaras
Environmental Research: Energy (Env. Research: Energy), 2025.
- Hardware Scaling Trends & Diminishing Returns in Large-Scale Distributed Training**
Jared Fernandez, Luca Wehrstedt, Leonid Shamis, Mostafa Elhoushi, Kalyan Saladi, Yonatan Bisk, Emma Strubell, Jacob Kahn
Transactions on Machine Learning Research (TMLR), 2025.
- Energy Considerations of Large Language Model Inference & Efficiency Optimizations**
Jared Fernandez*, Clara Na*, Vashisth Tiwari*, Yonatan Bisk, Sasha Luccioni, Emma Strubell
The 63rd Annual Meeting of the Association for Computational Linguistics (ACL), 2025.
- Holistically Evaluating the Environmental Impact of Creating Language Models**
Jacob Morrison, Clara Na, **Jared Fernandez**, Tim Dettmers, Emma Strubell, Jesse Dodge
International Conference on Learning Representations (ICLR), 2025.
- Gradient Localization Improves Lifelong Pretraining of Language Models**
Jared Fernandez, Yonatan Bisk, Emma Strubell
Findings of the ACL: EMNLP (EMNLP Findings), 2024.

7. **The Framework Tax: Disparities Between Inference Efficiency in Research & Deployment**

Jared Fernandez, Jacob Kahn, Clara Na, Yonatan Bisk, Emma Strubell
Workshop on Efficient Systems for Foundation Models at ICML (ES-FOMO), 2023.
Empirical Methods in Natural Language Processing (EMNLP), 2023.

8. **Generative Data Augmentation for Commonsense Reasoning**

Yiben Yang, Chaitanya Malaviya, Jared Fernandez, Swabha Swayamdipta, Ronan Le Bras, Ji-Ping Wang,
Chandra Bhagavatula, Yejin Choi and Doug Downey
Findings of the ACL: EMNLP (EMNLP Findings), 2020.

9. **VecShare: A Framework for Sharing Word Representation Vectors**

Jared Fernandez, Zhaocheng Yu, and Doug Downey
Empirical Methods in Natural Language Processing (EMNLP), 2017.

WORKSHOPS AND PREPRINTS

1. **Data Augmentation for Improving Contextual Faithfulness Under Knowledge Conflict**

Mohith Rajesh, Raj Maheshwari, Vashisth Tiwari, Yonatan Bisk, Emma Strubell, Jeremiah Milbauer, Jared Fernandez
Under Review at ACL Rolling Review (ACL ARR), 2026.

2. **AI x Crypto, Crypto x AI: A Survey**

Sarah Allen, Maryam Bahrani, Samuel Breckenridge, Aaron Buchwald, Christian Cachin, Jared Fernandez, James
Hsin-yu Chiang, Marwa Mouallem, Roi Bar-Zur, Ittay Eyal, Giulia Fanti, Ari Juels, Andrew Miller, Christian
Sillaber, Dani Vilardell, Matt Weinberg, Sen Yang, and Fan Zhang
Preprint (arXiv), 2026.

3. **The Energy Cost of Execution-Idle in GPU Clusters**

Yiran Lei, Jared Fernandez, Vasileios Kypriotis, Dimitrios Skarlatos, Emma Strubell, Justine Sherry, Daniel Vosler
Under Review at Symposium on Operating Systems Principles (SOSP), 2026.

4. **Evaluating the Environmental Impact of Language Models with Life Cycle Assessment**

Jared Fernandez*, Clara Na*, Yonatan Bisk, Emma Strubell
Best Proposal Award, Climate Change AI Workshop (CCAI at ICLR), 2025.

5. **Adapting to Gradual Distribution Shifts with Continual Weight Averaging**

Jared Fernandez, Saujas Vaduguru, Sanket Vaibhav Mehta, Emma Strubell
Workshop on High-Dimensional Learning Dynamics at ICML (HiDL), 2025.

6. **Efficiency Pentathlon: A Standardized Arena for Efficiency Evaluation**

Hao Peng, Qingqing Cao, Jesse Dodge, Matthew E. Peters, Jared Fernandez, Tom Sherborne, Kyle Lo, Sam
Skjonsberg, Emma Strubell, Darrell Plessas, Iz Beltagy, Evan Pete Walsh, Noah A. Smith, Hannaneh Hajishirzi
Preprint (Arxiv), 2023.

7. **Adapting to Gradual Distribution Shifts with Continual Weight Averaging**

Jared Fernandez, Saujas Vadaguru, Sanket Vaibhav Mehta, Yonatan Bisk, Emma Strubell
Workshop on High-Dimensional Learning Dynamics at ICML (HiDL), 2023.

8. **CIGLI: Conditional Image Generation from Language & Image**

Xiaopeng Lu, Lynette Ng, Jared Fernandez, Hao Zhu
Workshop on Closing the Loop Between Vision and Language at ICCV (CLVL), 2021.

9. **CODAH: An Adversarially Authored Question-Answer Dataset for Common Sense**

Michael Chen, Mike D’Arcy, Alisa Liu, Jared Fernandez, and Doug Downey
Workshop on Evaluating Vector Space Representations for NLP (RepEval), 2019.

10. **Sampling Informative Training Data for RNN Language Models**

Jared Fernandez and Doug Downey.
Association of Computational Linguistics: Student Research Workshop (ACL-SRW), 2018.

INVITED TALKS

1. **Energy Considerations in Efficient Language Model Development and Deployment**
Seagate Technology October 2025
Bosch Research June 2025
2. **Sampling Informative Training Data for RNN Language Models**
Midwest Speech and Language Days May 2018

TEACHING EXPERIENCE

On-Device Machine Learning with Profs. Yonatan Bisk & Emma Strubell	Fall 2023, 2024
Machine Learning with Prof. Doug Downey	Spring 2018
Discrete Math with Prof. Aravindan Vijayaraghavan	Fall 2017
Artificial Intelligence with Prof. Michael Rubenstein	Spring 2017

MENTORSHIP

Xiang Fan , <i>CMU BS → PhD at University of Washington</i>	2022 - 2023
Vashisth Tiwari , <i>CMU MS → PhD at CMU</i>	2025
Mohith Rajesh , <i>CMU MS</i>	Current
Raj Maheshwari , <i>CMU MS</i>	Current

DEPARTMENTAL SERVICE

PhD Admissions Committee , <i>Carnegie Mellon University</i>	2025
Graduate Application Support Program Mentor , <i>Carnegie Mellon University</i>	2020 – 2022
Undergraduate & Graduate Student Mentor , <i>Carnegie Mellon University</i>	2020 – 2022
Undergraduate Mentor , <i>Northwestern Eta Kappa Nu</i>	2019
External Vice President , <i>Northwestern IEEE Student Chapter</i>	2018
Technical Program Director , <i>Northwestern IEEE Student Chapter</i>	2017

PROFESSIONAL SERVICE

Reviewing	
Joule	2025 – 2026
COLM	2026
ACL Rolling Review	2024 – 2026
NeurIPS	2022, 2024 – 2026
ICLR	2022, 2024
ICML	2023
ACM Multimedia	2022
Student Volunteer	
EMNLP	2017
ACL	2018